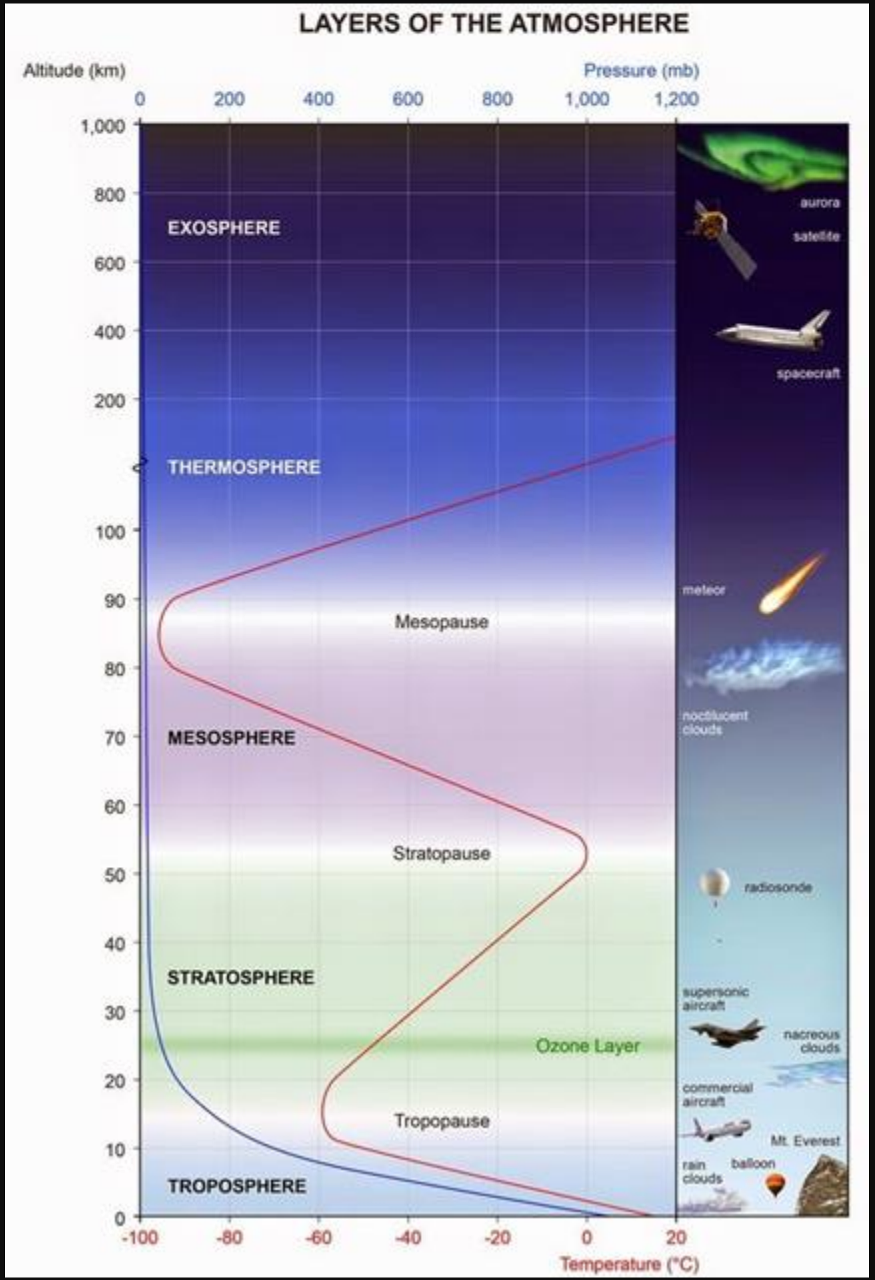
Greenhouse Gases, Good or Bad?

By: Valentina Sánchez-Castaño

The Earth's Atmosphere

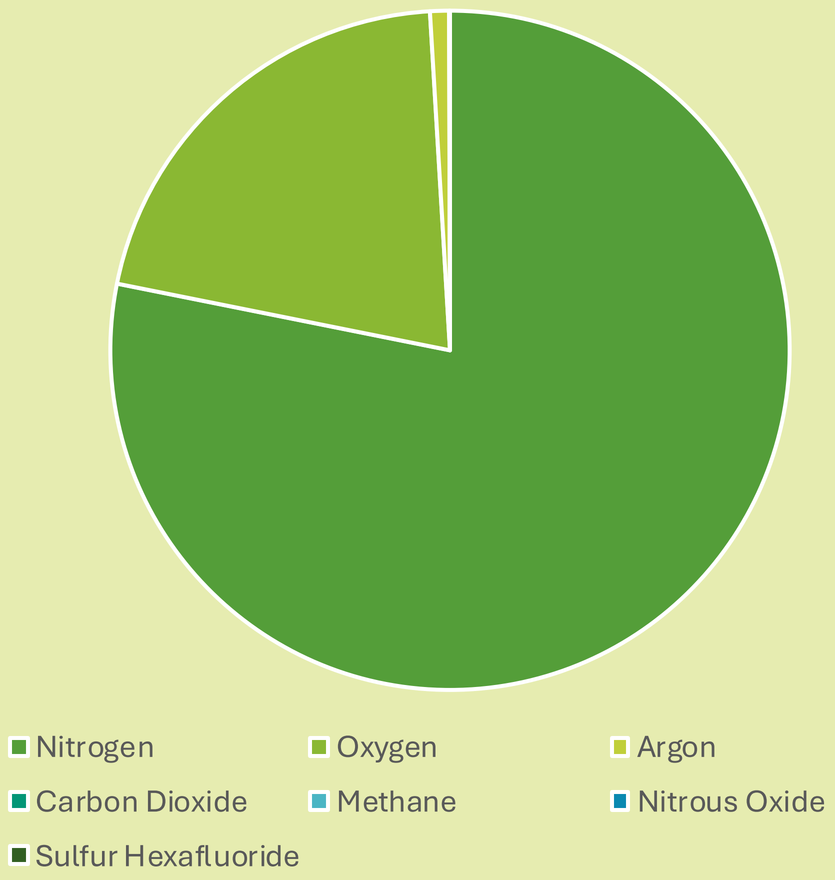
The Earth's atmosphere consists of five primary layers classified by temperature and density: **the troposphere, stratosphere, mesosphere, thermosphere, and exosphere.** These layers act as a protective shield, regulating energy balance, providing breathable air, and shielding life from solar radiation and meteors



The Earth's Atmosphere

Although **99.5%** of the atmosphere consists of **nitrogen, oxygen, and argon,** which do not absorb radiation, a tiny **0.43%** of "trace" gases like **water vapor** are responsible for trapping the infrared heat emitted by the Earth. Known as Greenhouse Gases (GHGs), these molecules act as a natural "blanket" for the planet. This critical process prevents heat from escaping directly into space and keeps Earth above the freezing point of water, making life possible.

Composition of Earth's Dry Atmosphere



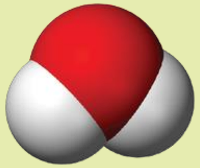
We cannot even see the other gases in the plot!!!!

BUT

Carbon dioxide (the second most important GHG), methane, nitrous oxide, sulfur hexafluoride, ozone, and a few others, in their current amounts, are sufficient to absorb a major fraction of the infrared light in the atmosphere.

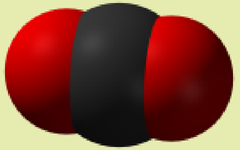
Let's see why... But first, let's see their composition

Water Vapor



The most abundant greenhouse gas, making up about 0.39% of the atmosphere and contributing significantly to natural warming.

Carbon Dioxide

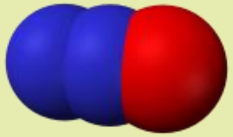


Produced by burning fossil fuels and organic matter; it is cycled through sinks like forests and oceans.

Methane

A highly combustible gas released from wetlands, landfills, and livestock digestive processes.

Nitrous Oxide



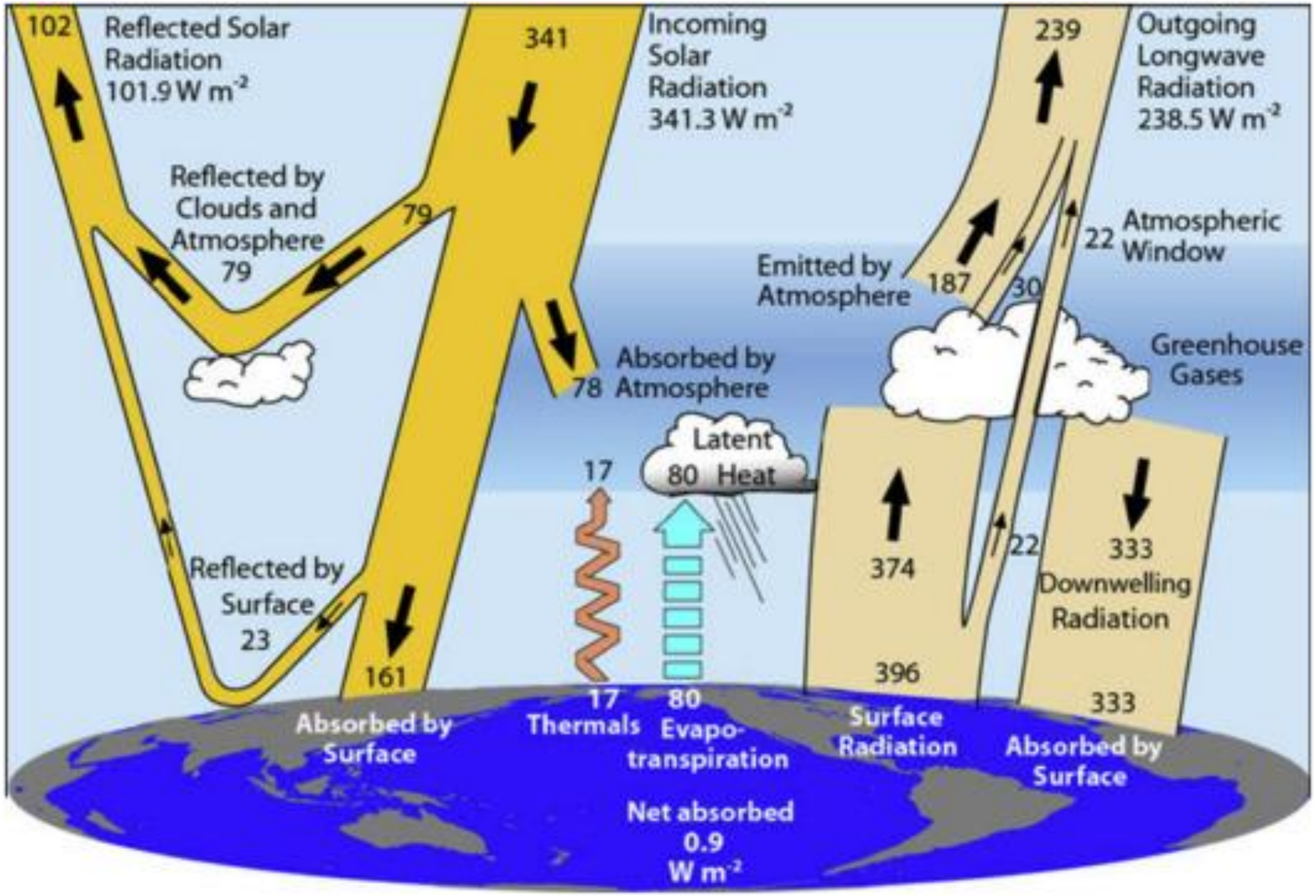
Often called laughing gas, it is released through industrial manufacturing, fertilizers, and natural rainforest processes.

Sulfur Hexafluoride

An extremely potent, human-made gas used in the power industry that persists in the atmosphere for over 1000 years.

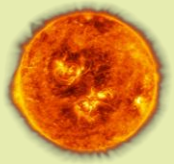
The Greenhouse Effect

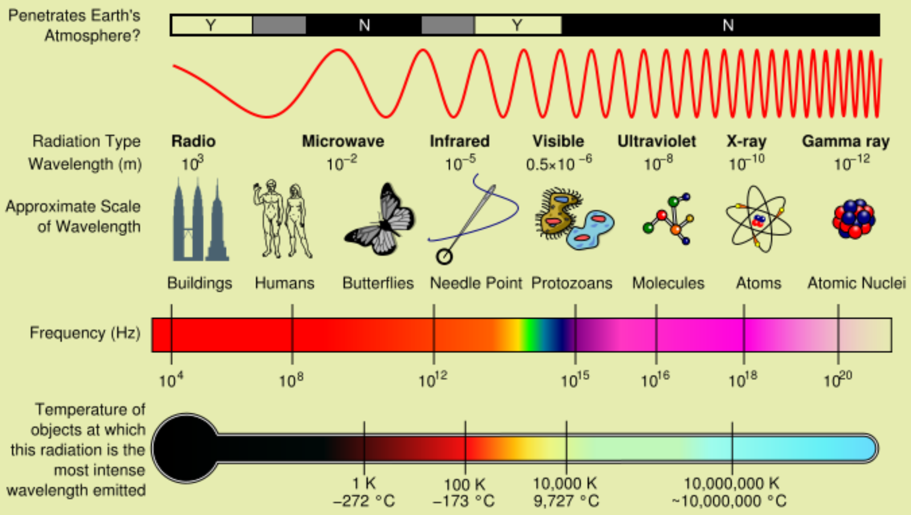
Sunlight, primarily in the form of shortwave radiation, is the **fundamental driver** of Earth's climate. While roughly 30% of this incoming energy is reflected back to space by clouds and the surface, and 20% is absorbed by the atmosphere, the remaining 50% is absorbed at the Earth's surface, warming the planet. Despite their small numbers, trace gases regulate this warmth through the Greenhouse Effect, which acts as a **natural blanket** that maintains temperatures suitable for life.



The Invisible Blanket

The Sun emits Shortwave Radiation, while the Earth tries to emit Longwave Radiation back into space.





250°F (121°C)
in daylight

Moon

-208°F (-133°C)
in nightfall

59°F (15°C)

If 100% of that heat were to escape, Earth would be a frozen rock.

But, we have an "invisible blanket" made of specific gases that trap that heat.

Why Do They Trap Heat?

The air is mostly Nitrogen and Oxygen. These are "stiff" molecules made of two identical atoms. Because they are perfectly balanced (symmetric), they don't have an electrical "handle" for the infrared light to grab onto. Infrared radiation passes right through them as if they weren’t even there. Greenhouse gases differ because they are more complex and asymmetric. When they bend or stretch, they create an oscillating dipole, essentially an electrical antenna. When the frequency of the Earth's infrared heat matches the natural "wiggle" frequency of the molecule, its "catches" the energy and starts dancing (vibrating). This is called Resonance.

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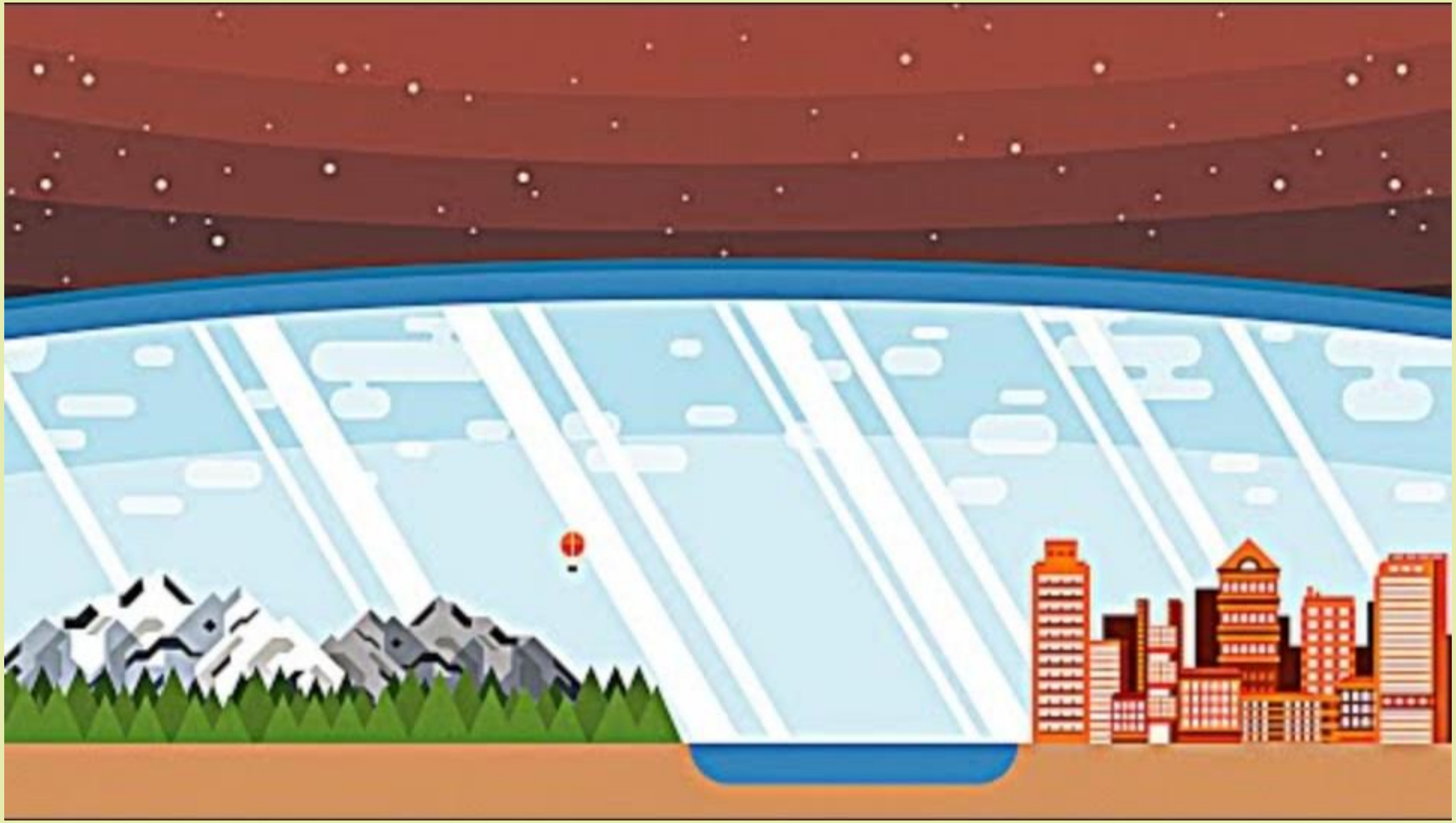
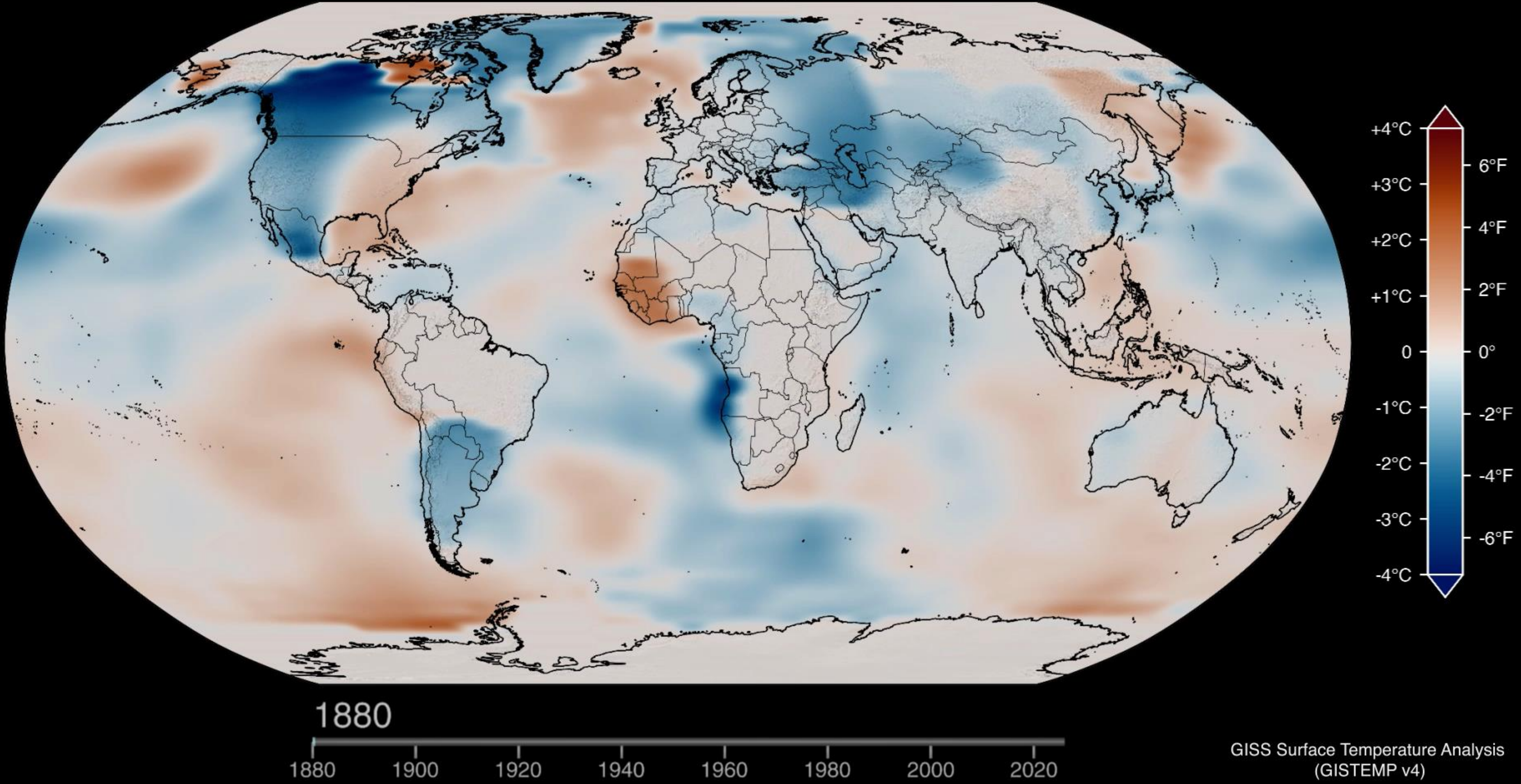


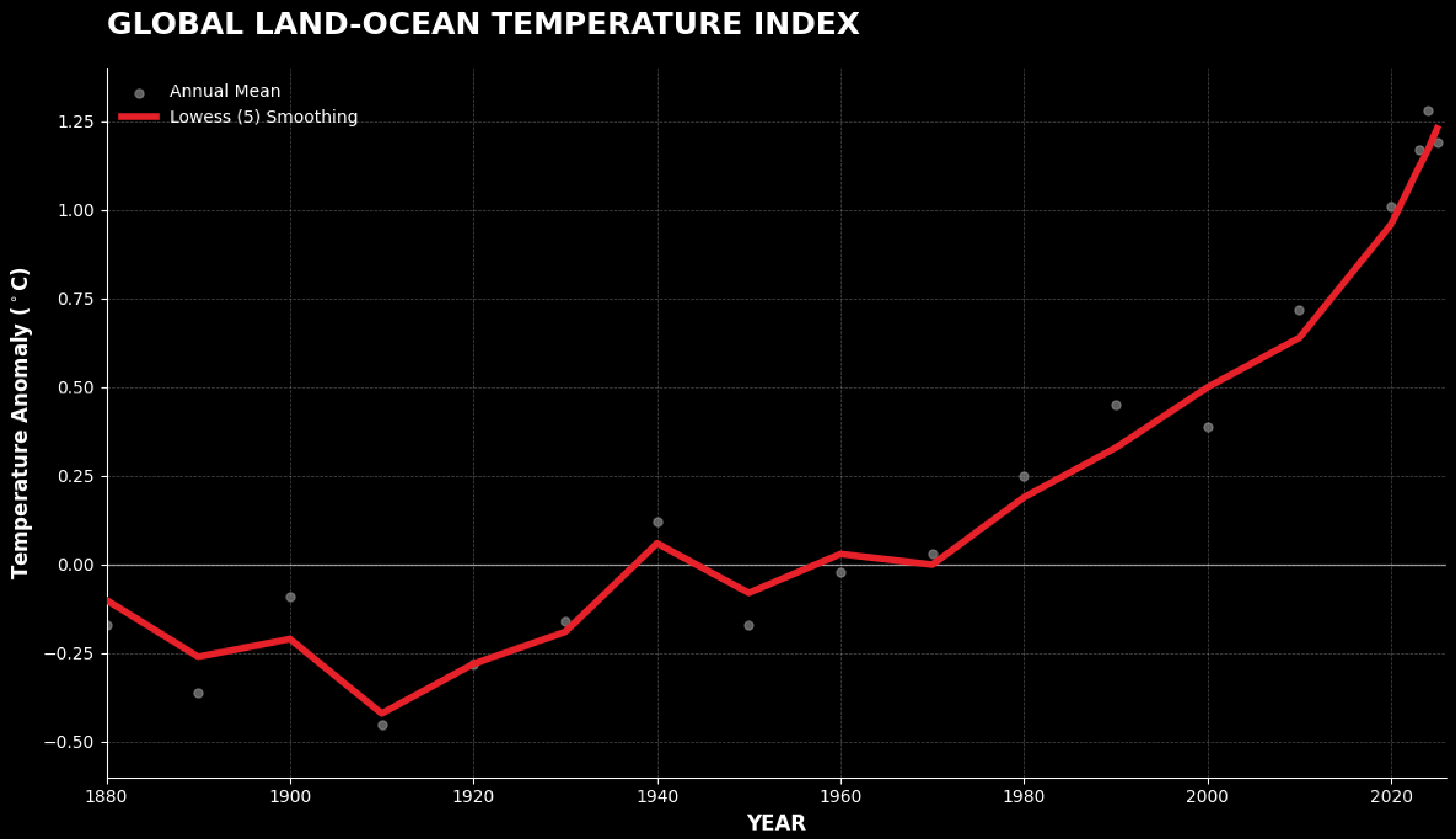
HOWEVER

Change in Global Temperature

relative to 1951-1980



GISS Surface Temperature Analysis
(GISTEMP v4)


GLOBAL LAND-OCEAN TEMPERATURE INDEX
Annual Mean
Lowess (5) Smoothing
Temperature Anomaly (°C)
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
1880
1900
1920
1940
1960
1980
2000
2020
YEAR

Discussion:

1. What is the best way that you can come up with for a country to decrease its greenhouse gas emissions?

2. If you were a policymaker trying to implement your solution, what objections do you think people would give to try to stop it?

3. How would you respond to convince them to implement the solution?

Thank You!!!